

✓ 100% Guaranteed Job Opportunities

IN COLLABORATION WITH

Indian Institute of Technology, Patna

 Technology Innovation Hub (TIH) · IIT Patna

COURSE BROCHURE

Executive PG in DevOps & Multi-Cloud with Gen AI Job Ready Program

COURSE CURRICULUM · TIH-IIT PATNA CERTIFIED

250+

TRAINING HOURS

5

CLOUD TRACKS

20+

PROJECTS

5

CERTIFICATIONS

PARTNERS & CERTIFICATIONS



TIH-IIT Patna



AWS Partner



Microsoft

5/5 ★ Google

Live

ONLINE TRAINING

Lifetime

RECORDING ACCESS

PPT+Labs

STUDY MATERIAL

5/5 ★

GOOGLE RATING

PROGRAMME STRUCTURE

Course Curriculum

5 tracks · 250+ hours · 20+ projects · Hands-on labs

Linux Fundamentals

10 hrs

- ▶ Distributions: Ubuntu, CentOS, Amazon Linux; CLI vs GUI; Kernel / Shell / User Space
- ▶ FHS; Boot process; Directory structure; File types; Absolute vs relative paths
- ▶ File & directory management; File permissions & ownership; wildcards
- ▶ Users & Groups; sudo; /etc/passwd & /etc/shadow
- ▶ Process management; Package management: apt, yum/dnf; Installing Git, Docker, Java
- ▶ Networking basics; Disk management; Text processing: grep, awk, sed, sort, cut
- ▶ Archiving: tar, gzip; systemctl (Apache & Nginx); Environment variables; Alias

Shell Scripting

8 hrs

- ▶ Shell types: bash, sh, zsh; Writing first script; Shebang; Running scripts
- ▶ Variables & input; Operators: Arithmetic, Relational, Logical
- ▶ Conditionals: if, if-else, elif, case; Loops: for, while, until
- ▶ Functions: creating & passing arguments; File handling; Log processing
- ▶ Cron scheduling; CI/CD script automation; Jenkins & Docker integration

Python Scripting

8 hrs

- ▶ Python for DevOps: automation, APIs, cloud use cases
- ▶ Variables, data types, I/O, type casting, if-else, loops
- ▶ Functions; Data structures: Lists, Tuples, Sets, Dicts
- ▶ File handling; Exception handling: try-except, finally
- ▶ os module; subprocess module; boto3; AWS task automation; venv & pip

Git & GitHub

10 hrs

- ▶ VCS: Centralized vs Distributed; Working Dir / Staging / Repository
- ▶ Git objects; Lifecycle (tracked/untracked); SHA-1
- ▶ Install; SSH vs HTTPS; Init, add, commit, log, diff
- ▶ Branching: Git Flow vs Trunk-Based; Merge vs Rebase; Conflict resolution

- ▶ Pull Requests; Code reviews; Branch protection; Forking vs cloning
- ▶ Stash; Cherry-pick; Reset vs Revert; Tagging; .gitignore; Rollback

Maven + Tomcat

4 hrs

- ▶ Maven: build tools overview; Installation on Linux
- ▶ pom.xml; Dependencies; Maven repos; Goals & terminology
- ▶ Artifacts: JAR/WAR; Intro to Nexus/JFrog
- ▶ Tomcat: directory structure, config files, port change
- ▶ Manual Java Web App deployment → need for automation

Jenkins + Nexus + SonarQube

14 hrs

- ▶ CI/CD types; Jenkins setup on Amazon Linux & Ubuntu
- ▶ Freestyle & Declarative (Jenkinsfile) pipelines; Blue Ocean
- ▶ Triggers: Webhooks, Poll SCM, Build Periodically, Remotely
- ▶ Artifacts: Nexus, S3; Email & Slack notifications; Parallel stages
- ▶ Master/Worker architecture; RBAC; Multi-Branch Pipeline; Backup to S3
- ▶ Nexus artifact repository; SonarQube: quality gates & analysis

Jenkins Final Project

Jenkins + GitHub + Maven + Nexus + SonarQube + S3 + Tomcat + Email · Hotstar Clone App

Docker

14 hrs

- ▶ Monolith vs Microservices; Docker architecture & workflow
- ▶ Dockerfile: custom images; Dockerize HTML, Java, Spring Boot, Python
- ▶ Multi-stage builds; Image layers & caching; Non-root users
- ▶ Push to DockerHub, AWS ECR, GitHub Container Registry
- ▶ Volumes; Docker-Compose 3-Tier App; Networks; Swarm overview
- ▶ Trivy scanning; Docker Scout; Secrets best practices

Docker Final Project

Jenkins + Trivy + Docker – End-to-End CI/CD Pipeline

Kubernetes (EKS)

30 hrs

- ▶ K8s Architecture; EKS Cluster creation; Basic workloads overview
- ▶ Pods: types, multi-container, init, static, restart policies, troubleshooting
- ▶ Services: ClusterIP, NodePort, LB; Namespaces; RC, RS, Deployments
- ▶ Strategies: Rolling, ReCreate, Blue-Green, Canary; DaemonSet; Scheduling
- ▶ ConfigMap & Secrets; AWS Secrets Manager; HPA; Cluster Autoscaler; EKS Auto-mode
- ▶ PV & PVC; StatefulSet; Ingress (AWS LB Controller); Jobs; CronJobs; RBAC, OIDC, IRSA
- ▶ ArgoCD; Helm custom chart (Zomato App); Prometheus; Grafana; EFK; Istio

Ansible

10 hrs

- ▶ Config management; Agentless SSH architecture; Inventory (static & dynamic)
- ▶ Ad-hoc commands; Core modules: file, copy, yum/apt, service, user
- ▶ Playbooks: YAML, tasks, handlers; Variables; Conditionals & loops
- ▶ Roles: structure & reusability; Templates & Jinja2; Ansible Vault

Terraform

10 hrs

- ▶ IaC: Declarative vs Imperative; Terraform vs CloudFormation/ARM
- ▶ Providers, Resources, State file, lifecycle; Configure AWS CLI
- ▶ init, plan, apply, destroy; EC2, S3; State locking (S3 + DynamoDB)
- ▶ Input variables; tfvars; Output values; Modules; Workspaces (dev/stage/prod)

Terraform Project

Infrastructure Deployment (Terraform) + App Deployment (Jenkins)

GitHub Actions

8 hrs

- ▶ Workflows, jobs, steps, runners & triggers
- ▶ CI/CD for Docker Deployment; Accessing AWS via GHA (OIDC)
- ▶ Variables; GitHub-hosted & self-hosted runners

GHA Projects

Dockerizing Java App · Dockerizing Python App

DevOps Capstone Projects

Capstone Project 1

3-Tier Microservices Architecture using Jenkins

Capstone Project 2

Microservices App using Jenkins – End-to-End Pipeline

Additional Projects

Zomato Clone App · BookMyShow Clone App · Microservices App

02 Amazon Web Services (AWS)



50 Hours

Fundamentals & IAM

- ▶ Cloud Computing; Global Infrastructure (Regions, AZs); Service categories
- ▶ IAM: Users, Groups, Roles, Policies (JSON), MFA; KMS, Secrets Manager, WAF, Security Hub

Compute, Storage & Databases

- ▶ EC2, Auto Scaling, ELB; Instance types & pricing; HA design
- ▶ S3, EBS, EFS, FSx, AWS Backup; Lifecycle strategies
- ▶ RDS (Multi-AZ, read replicas); DynamoDB (on-demand, indexes)

Networking & Monitoring

- ▶ VPC: subnets, routing, IGW, NAT Gateway; Secure networking
- ▶ Route 53 & ACM; Transit Gateway; Global Accelerator; Network Firewall
- ▶ CloudWatch, CloudTrail, AWS Config; Metrics, logs & tracing

Containers, Serverless & CI/CD

- ▶ ECS, EKS, Fargate, ECR; Lambda, API Gateway, Step Functions, EventBridge, SQS, SNS
- ▶ CodeCommit, CodeBuild, CodeDeploy, CodePipeline; CloudFormation; SSM

- ▶ Organizations, Control Tower; Cost Explorer, Budgets, Trusted Advisor

AWS Capstone Projects

Project 1

End-to-End CI/CD Pipeline (CodePipeline + EC2/ECS)

Project 2

Highly Available Architecture (ALB + Auto Scaling + Multi-AZ)

Project 3

Event-Driven Serverless System (Lambda + SQS + SNS + API GW)

Prepares for: AWS Solutions Architect Associate · AWS DevOps Professional · AWS Cloud Practitioner

03 Microsoft Azure + DevOps



45 Hours

Part A – Azure DevOps Platform (~23 hrs)

- ▶ Portal overview; Organizations, Projects & User Management; Work item processes (Agile/Scrum/CMMI)
- ▶ Boards: Kanban, Backlogs, Sprints; Repos: Git & TFVC; Branching strategies
- ▶ Pipelines: CI/CD; Classic Editor & YAML multi-stage; Agents, Artifacts, Environments, Stages
- ▶ Test Plans; YAML: parallel stages, conditions & dependencies; Deployment Jobs
- ▶ App Service (PaaS); Release Management; Docker Container Instances & ACR
- ▶ Terraform on Azure CI/CD; AKS: kubectl, pods, YAML deployments

Part B – Azure Administration AZ-104 (~22 hrs)

- ▶ ARM, Resource Groups, PowerShell management
- ▶ VMs: VMSS, Autoscaling, Availability Sets/Zones, Disk Encryption, Key Vault
- ▶ Storage: Blob (Page/Block/Append), File, Table, Queue; Storage Key management
- ▶ Networking: VNets, NSG, L4/L7 Load Balancers, Traffic Manager Profile routing

- ▶ Azure SQL: vCore/DTU; PaaS & App Services; Azure Monitor; Log Analytics Workspace
- ▶ Azure AD; RBAC roles; Subscriptions; Resource locks; PowerShell automation
- ▶ AI/Copilot for Azure: provision IaaS/PaaS/SaaS; automate operations & billing

Azure Capstone Projects

Project 1

Azure DevOps CI/CD Pipelines – E-Learning web application on Azure App Service

Project 2

High-availability architecture distributing traffic across VMs via Azure Load Balancer

Prepares for: Azure Administrator AZ-104 · Azure DevOps Engineer AZ-400

04 Google Cloud Platform (GCP)



30 Hours

Why GCP & Cloud Basics

- ▶ Traditional IT vs Cloud; SaaS/PaaS/IaaS; GCP locations; Gartner Magic Quadrant
- ▶ GCP Console, Cloud Shell, Organizations, Folders, Projects; Billing alerts

IAM, Compute Engine & GCS

- ▶ IAM: roles, members, service accounts, policies; resource hierarchy; best practices
- ▶ Compute Engine: VMs, lifecycle, snapshots, machine images, storage types
- ▶ GCS: buckets, IAM/ACL, versioning, lifecycle policies
- ▶ VPC: Auto/Custom mode, peering, NAT, Firewall; Hybrid: VPN & Interconnect
- ▶ Load Balancing: global/regional, internal/external; MIGs; Autoscaling

Databases & Serverless

- ▶ Cloud SQL: MySQL, PostgreSQL, SQL Server; HA, replication, backup
- ▶ Cloud Spanner: global DB; Firestore/Datastore: document model; Bigtable: big data
- ▶ App Engine: Standard vs Flexible; scaling types; Labs

- ▶ Cloud Functions & Cloud Run; Cloud Monitoring & Logging

GKE & Cert Prep

- ▶ GKE: Microservices, Containers, Docker Images, Container Registry
- ▶ Kubernetes: architecture, nodes & clusters; Pods, Services, RC; YAML Deployments
- ▶ Labs: Create K8s Cluster; Deploy WordPress; Manage Pods

GCP Capstone Projects

Project 1 – VPC & Compute Engine

Auto Mode & Custom VPC · Subnets · VMs · Firewall Rules · VPC Peering · NAT Gateway

Project 2 – 2-Tier App in Kubernetes

Guestbook App (Frontend + Redis Master & Slave) · Scale Up/Down · Upgrade & Downgrade Pods

Prepares for: GCP Professional Cloud Architect · GCP Associate Cloud Engineer · GCP Cloud Digital Leader

05

AI for DevOps & MLOps

32 Hours

Foundations & ML Pipeline Basics

4 hrs

- ▶ AI/ML/Deep Learning concepts; DevOps vs MLOps lifecycle comparison
- ▶ ML challenges: data drift, reproducibility, deployment issues
- ▶ Python ML pipeline: sklearn; train.py & predict.py structure

Git for ML, DVC & MLflow

7 hrs

- ▶ Git for ML: repo setup, branching & collaboration workflow
- ▶ DVC: data versioning, pipeline stages (data → preprocess → train); dvc repro
- ▶ MLflow: logging metrics, parameters, artifacts; Model registry; Compare & register best model

Model Serving & Containerisation

4 hrs

- ▶ FastAPI: prediction API backend; Streamlit: UI dashboard
- ▶ Dockerizing ML APIs: Dockerfile, build & run container
- ▶ Kubernetes: deploy FastAPI ML app & expose service

CI/CD, Kubeflow & Monitoring

11 hrs

- ▶ GitHub Actions CI/CD: Data validation → Training → Evaluation → Deploy
- ▶ Secrets management; Conditional deployment; Self-hosted runners
- ▶ Kubeflow: pipeline orchestration, components & workflow tracking
- ▶ Prometheus: custom metrics on FastAPI ML app; Grafana: dashboards
- ▶ AWS SageMaker: Studio, Training Jobs, Endpoints, Model Registry, Pipelines

Capstone: Production MLOps Pipeline

2 hrs

End-to-End Production MLOps

Data → Preprocess → Train → Serve → Monitor | GitHub Actions CI/CD | Docker on Kubernetes | Prometheus + Grafana

WHAT YOU EARN

Certifications You Will Earn



TIH-IIT Patna DevOps Certification

Technology Innovation Hub — IIT Patna

Globally recognised. Stand out with an IIT-backed credential.



AWS Solutions Architect Certificate

Amazon Web Services



Azure Administrator AZ-104

Microsoft Azure



DevOps AWS Azure GCP Gen AI Expert

Paperlive Learning



Project Experience Certification

Paperlive Learning — Verified Portfolio

SAMPLE CERTIFICATIONS

Credentials you take home

Five certificates verify your training, hands-on project work, and globally recognized cloud expertise.

01

TIH-IIT Patna Certificate of Completion

Issued in collaboration with IIT Patna i-Hub Foundation



IIT Patna Vishlesan i-HUB
Foundation



Offline Training Center Name
Centre ID: EDZ-XX-XXX-XX



Edzor Centre of AI &
Technological Excellence

CERTIFICATE OF COMPLETION

This is to certify that

{Student Name}

has successfully completed the

{Program Name},

organized by Edzor Centre of AI & Technological Excellence,

in collaboration with **IIT Patna Vishlesan i-HUB Foundation,**

at the offline upskilling centre administered by

{Training Centre Name}.

[Authorized Signatory Name]

[Designation],

IIT Patna Vishlesan i-HUB Foundation



Certificate ID : EDZ/XXXX/XXXXX

Issue Date : XX-XXX-XXXX

4-JUNE-2025



In collaboration with

A circular seal with a dark blue background and a red border, containing the text 'CERTIFICATE OF COMPLETION' in white.

**CERTIFICATE
OF COMPLETION**

PROUDLY CERTIFIES THAT

RAKSHA

has successfully finished all training modules,
practical projects, and assessments under the
direction of competent mentors and instructors at
PaperLive Learning, receiving a Course Completion
Certificate in **DevOps, AWS, Azure and GCP.**

A handwritten signature in black ink, reading 'Adil Hasnain'.

ADIL HASNAIN

CO-FOUNDER

A handwritten signature in black ink, reading 'Sayed Azmat Ali'.

SAYED AZMAT ALI

CO-FOUNDER

CERTIFICATION CODE

PL-053-10

aws  certified

ETHAMUKKALA KARTHIK

AWS Certified Solutions Architect - Associate

VALIDATION NUMBER: 85bf2c5a136f4158b61dbc940c6c9cb4

VALIDATE AT: <https://aws.amazon.com/verification>

Issue Date: April 3, 2026

Expiration Date: April 3, 2029

4/14/25, 7:47 PM

Credentials - sayed-1521 | Microsoft Learn





ADAZ PAPERLIVE LEARNING LLP

support@paperlive.in

Project Experience Certificate

Certificate No.: [PL-053-10]

This is to certify that **Raksha** has successfully completed hands-on project work as part of the **DevOps and Cloud Job-Ready Program at PaperLive Learning**, conducted from **December 2024 - May 2025**.

During the program, the candidate demonstrated practical knowledge and hands-on expertise in executing projects involving:

- DevOps Tools: Jenkins, Docker, Kubernetes, Git, GitHub, Terraform, Ansible
- Cloud Platforms: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- CI/CD Pipelines: Implementation of continuous integration and deployment workflows
- Infrastructure Automation: Using Infrastructure as Code (IaC) practices
- Monitoring & Deployment: Hands-on deployment, scaling, and monitoring of cloud-based applications

Project Areas Covered:

- CI/CD pipeline automation for microservices-based applications
- Infrastructure provisioning and deployment on AWS & Azure using Terraform
- Container orchestration using Kubernetes
- Cloud monitoring and security implementation
- Capstone project simulating enterprise-level DevOps lifecycle

The candidate has shown dedication and a strong understanding of cloud-native development and DevOps practices, making them well-prepared for industry roles in this domain.

We wish them continued success in their professional journey.

Sincerely,

Sayed Azmat Ali
Adil Hasnain
CEO
Paperlive Learning

www.paperlive.in

EXCLUSIVE BENEFITS

TIH-IIT Patna Exclusive Benefits

Vishlesan I-Hub Foundation · IIT Patna





Official Student ID Card & Admission Letter from TIH-IIT Patna



TIH-IIT Patna Admission Kit – physical welcome kit from the institute



6+ Guest Lectures from TIH-IIT Faculty



TIH-IIT Patna DevOps Certification – globally recognized credential

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5000+ Companies Hiring Our Alumni

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OUR INSTRUCTORS

Learn from Industry Experts

PK

Puneet Kumar Bhatia

Microsoft Certified Trainer

20 years of IT experience – Requirement Analysis, Product Development & Testing, DevOps, cloud computing, and Agile Methodologies.

RS

Ritvik Saxena

AWS Cloud & DevOps Trainer

Experienced AWS Cloud and DevOps Trainer dedicated to empowering learners with industry-ready skills through hands-on training in AWS, DevOps tools and automation.

TP

Tanvi Puri

Microsoft Certified Trainer

DevOps Enthusiast with expertise in automating software lifecycles using Azure DevOps, Docker, Kubernetes, and Ansible. Expert in designing CI/CD pipelines.



Aneesh Kumar

AWS & Azure Certified Expert

Certified IT expert in AWS and Azure, working as a DevOps Architect specialising in cloud and infrastructure for critical applications in banking and finance.



Arun Raju

Microsoft Certified Trainer

DevOps Engineer and Trainer specialising in Linux, Shell Scripting, Git, Jenkins, Docker, Terraform, and Ansible. Builds scalable, automated infrastructure.



Loksai

DevOps Architect & Trainer

Specialises in GIT, AWS DevOps, Azure DevOps, Kubernetes, ArgoCD, Docker, Terraform, Ansible, and Jenkins – streamlining CI/CD pipelines across enterprises.

PLACEMENT SUPPORT

Career Assistance Programme

- ✓ Role-based Resume Building
- ✓ LinkedIn Profiling & Optimisation
- ✓ 1000+ Interview Question Bank
- ✓ Soft Skills Grooming Sessions
- ✓ Real-time Workshop with Top MNCs
- ✓ Unlimited Mock Interviews

✓ 6 Guaranteed Job Opportunities Every Month

WHY CHOOSE US

Why Choose Paperlive Learning

250+

TRAINING HOURS

200+

HRS PRACTICAL

150+

MENTORSHIP

20

CAPSTONE PROJECTS

50+

CASE STUDIES

5000+

HIRING PARTNERS

6+

IIT GUEST LECTURES



Live Training from IIT Faculty

Learn directly from TIH-IIT Patna faculties – real academic rigour with industry depth.



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100% placement assistance backed by 5000+ hiring partners. We don't stop until you're placed.



Official AWS & Microsoft Certs

Earn AWS Solutions Architect and Microsoft AZ-104 – globally recognised credentials.



TIH-IIT Patna Certification

Official certification, admission letter, student ID & admission kit from IIT Patna.



Resume & Interview Prep

Dedicated sessions: resume building, mock interviews & certification exam preparation.



Lifetime Recording Access

Every live session recorded and available forever. Revisit any topic at any time.

LEARNING OUTCOMES

What You Will Be Able To Do

- > Design and deploy scalable, HA architectures on AWS, Azure & GCP
- > Build CI/CD pipelines with Jenkins, GitHub Actions & Azure DevOps
- > Containerise and orchestrate apps with Docker & Kubernetes at scale
- > Automate infrastructure across multi-cloud with Terraform & Ansible
- > Build AI/ML pipelines and MLOps workflows with DVC, MLflow & Kubeflow
- > Monitor cloud & ML workloads with Prometheus, Grafana & CloudWatch
- > Secure cloud resources with IAM, RBAC, KMS & policy frameworks
- > Clear 5 certifications: TIH-IIT Patna, AWS, Azure, DevOps Expert, Project Experience

STUDENT REVIEWS

What Our Students Say

5.0 ★★★★★



Anil Shrivastava

Hewlett Packard Enterprise



Really Changed Career!!

My career has completely changed since I enrolled in PaperLive Learning's DevOps and Cloud Training! The course has been a life-changing experience with exceptional mentorship.



Dilip Mehta

PhonePe



Absolutely Game Changer!!

The DevOps training I received from PaperLive Learning fundamentally changed the way I think about software development. The course created a solid production-ready foundation.



Suraj Sharma

Concentrix



Amazing Learning Experience!!

I thoroughly enjoyed this DevOps and Cloud course! The instructor provided exceptional support at every stage and was incredibly knowledgeable about the latest industry tools.



Zaid Hassmi

Deloitte



Highly Recommend This!!

I had a great time with this DevOps and Cloud course. The instructor was incredibly knowledgeable and provided excellent support throughout the entire learning journey.



Sridevi Rao

Cognizant



Really Appreciate This!!

This training was comprehensive and practical. We are really grateful to the instructor for their exceptional dedication to making us industry-ready professionals.



Sameer Suman

CRED



Helped Me Grow!!

PaperLive provides great assistance — one-on-one mentoring sessions and a variety of code tasks to help close the knowledge gap and build real production skills.



Ayush Raj

Accenture



Helped Me Transition!!

Originally from a non-tech background, PaperLive's outstanding one-on-one mentorship sessions made the transition to DevOps completely seamless and achievable.



Apurva Raj

Capgemini



Helped Me Succeed!!

After completing the Cloud and Job Ready Programme at PaperLive, securing a job in AWS and DevOps became straightforward. Highly recommend to anyone looking to transition.



Farzil Hai

Qualcomm



Learned Valuable Skills!!

After finishing PaperLive's Cloud and Job Ready Programme, I got placed at Qualcomm. The hands-on curriculum and dedicated placement support made all the difference.

[See All Reviews on Google →](#)

START YOUR JOURNEY TODAY

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